VIVEKANANDHA SAINIK COACHING CENTER

ALL INDIA SAINIK SCHOOLS ENTRANCE EXAMINATION (AISSEE) 2026

CLASS VI MOCK QUESTION PAPER 09

Candidate Name: _____ **Roll No:** _____

Time Allowed: 150 Minutes (2½ Hours) | Maximum Marks: 300 Marks

GENERAL INSTRUCTIONS FOR CANDIDATES:

1. This paper consists of **125 Multiple Choice Questions** divided into 4 Sections: Language (Q.1-25), Mathematics (Q.26-75), Intelligence (Q.76-100), and General Knowledge (Q.101-125).
2. Questions in Mathematics carry **3 Marks** each. Questions in Language, Intelligence, and General Knowledge carry **2 Marks** each.
3. There is **NO Negative Marking** for wrong answers.
4. Mark your answers clearly on the OMR Answer Sheet provided using a Blue/Black Ballpoint pen only.

SECTION A — LANGUAGE (ENGLISH) | 25 Questions x 2 Marks = 50 Marks

Read the following passage carefully and answer the questions that follow (Q.1 to Q.5):

Conservation of forests and wildlife is essential to maintain ecological balance on Earth. Forests are often called the 'green lungs' of our planet because they absorb harmful carbon dioxide and release life-giving oxygen through photosynthesis. They also prevent soil erosion, regulate rainwater flow, and provide natural habitat to millions of animal and bird species.

Deforestation—the large-scale clearing of trees for industrialization and urban expansion—has led to severe environmental crises such as global warming, unpredictable weather patterns, and loss of biodiversity. Programs like 'Van Mahotsav' encourage citizens to plant trees and protect existing green spaces. It is every individual's sacred duty to cherish and preserve our natural heritage for future generations.

Q.1. Why are forests described as the 'green lungs' of Earth?

A) Because they produce timber for building houses

B) Because they absorb carbon dioxide and release oxygen

C) Because they cover one-third of Earth's landmass

D) Because they attract tourists from across the world

Q.2. What major environmental crisis is caused by deforestation?

A) Increase in biodiversity

B) Global warming and loss of biodiversity

C) Abundant rainfall throughout the year

D) Enrichment of soil minerals

Q.3. Which annual initiative mentioned in the passage encourages tree plantation?

A) Swachh Bharat Abhiyan

B) Van Mahotsav

C) Jal Shakti Abhiyan

D) Green Revolution

Q.4. How do trees protect the topsoil?

A) By absorbing rainwater completely
B) By preventing soil erosion
C) By converting soil into sand
D) By blocking sunlight

Q.5. Which word in the passage is an antonym for 'DESTRUCTION'?

A) Biodiversity
B) Conservation
C) Erosion
D) Expansion

Q.6. Identify the proper noun in: 'Suraj bought a new bicycle last Monday.'

A) Bicycle
B) Suraj
C) Bought
D) New

Q.7. Fill in the blank with the correct possessive pronoun: 'This trophy belongs to us; it is _____.'

A) ours
B) our
C) us
D) ours'

Q.8. Select the correct comparative degree: 'Gold is _____ than silver.'

A) costly
B) more costly
C) most costly
D) costlier

Q.9. Choose the correct article: 'She has been waiting for _____ hour at the bus station.'

A) a
B) an
C) the
D) no article

Q.10. Choose the correct preposition: 'The sun disappeared _____ the clouds.'

A) behind
B) under
C) inside
D) between

Q.11. Choose the correct conjunction: 'You will not succeed _____ you work consistently.'

A) unless
B) if
C) until
D) because

Q.12. Choose the synonym of DILIGENT.

A) Lazy
B) Hardworking
C) Careless
D) Dishonest

Q.13. Choose the antonym of 'BENEFICIAL':

A) Useful
B) Advantageous
C) Harmful
D) Helpful

Q.14. Identify the correctly spelled word:

A) Necessary
B) Necesary
C) Neccessary
D) Necessery

Q.15. Give one-word substitution for: 'A life history of a person written by himself'

A) Biography
B) Autobiography
C) Bibliography
D) History

Q.16. Select the correct plural form of 'Ox':

A) Oxes
B) Oxen
C) Oxs
D) Oxies

Q.17. Select the correct masculine gender of 'Niece':

A) Uncle
B) Nephew
C) Cousin
D) Brother

Q.18. Choose the correct collective noun: 'A _____ of stars illuminated the night sky.'

A) galaxy
B) fleet
C) bunch
D) team

Q.19. Identify the sentence written in Past Continuous Tense:

A) She plays the guitar.
B) She played the guitar.
C) She was playing the guitar.
D) She has played the guitar.

Q.20. Fill in the blank: 'Neither Rohit nor his friends _____ present in the auditorium.'

A) was
B) were
C) is
D) has

Q.21. Identify the abstract noun: 'Wisdom is better than riches.'

A) Wisdom
B) Riches
C) Better
D) Than

Q.22. Change into Active Voice: 'The letter was delivered by the postman.'

A) The postman delivers the letter.
B) The postman delivered the letter.
C) The postman has delivered the letter.
D) The postman was delivering the letter.

Q.23. Complete the proverb: 'Actions speak louder than _____.'

A) promises
B) words
C) thoughts
D) deeds

Q.24. Choose the correct meaning of idiom: 'To spill the beans'

A) To drop food accidently
B) To reveal a secret prematurely
C) To cook a delicious meal
D) To waste money foolishly

Q.25. Rearrange into a meaningful sentence: 'rises / early / in the morning / my grandfather'

A) My grandfather early rises in the morning.
B) My grandfather rises early in the morning.
C) In the morning my grandfather early rises.
D) Rises early my grandfather in the morning.

SECTION B — MATHEMATICS | 50 Questions x 3 Marks = 150 Marks

Q.26. Find the face value of digit 8 in the number 9,87,654.

A) 80,000
B) 8,000
C) 800
D) 8

Q.27. Find the sum of the place values of two 5s in the number 5,45,210.

A) 5,05,000
B) 5,00,500
C) 5,50,000
D) 5,00,050

Q.28. Convert Hindu-Arabic numeral 948 into Roman numerals.

A) CMXLVIII
B) DCCCCXLVIII
C) CMXLXVIII
D) MCXLVIII

Q.29. Find the estimated difference between 8,725 and 3,480 by rounding off to the nearest thousand.

A) 5,000
B) 6,000
C) 5,200
D) 5,500

Q.30. Simplify using BODMAS rule: 72 / [12 + {18 - (6 x 2)}]

A) 2
B) 3
C) 4
D) 6

Q.31. Find the greatest 4-digit number which is exactly divisible by 15, 20, and 25.

A) 9,900
B) 9,600
C) 9,750
D) 9,850

Q.32. Find the HCF of two distinct prime number a and b.

A) a x b
B) a + b
C) 1
D) 0

Q.33. Simplify: 5 (1/3) - 2 (1/4) + 1 (1/2)

A) 4 (7/12)
B) 4 (5/12)
C) 4 (1/12)
D) 5 (1/12)

Q.34. Arrange the following decimals in descending order: 0.45, 0.54, 0.054, 0.405

A) 0.54 > 0.45 > 0.405 > 0.054
B) 0.54 > 0.405 > 0.45 > 0.054
C) 0.45 > 0.54 > 0.405 > 0.054
D) 0.054 > 0.405 > 0.45 > 0.54

Q.35. Multiply 3.45 by 1.2.

A) 4.14
B) 4.24
C) 4.04
D) 3.94

Q.36. A train 200 m long crosses a pole in 10 seconds. What is the speed of the train in km/h?

A) 60 km/h
B) 72 km/h
C) 80 km/h
D) 90 km/h

Q.37. If 18 workers can construct a wall in 15 days, how many workers are needed to finish it in 10 days?

A) 24 workers
B) 27 workers
C) 30 workers
D) 32 workers

Q.38. At what rate of simple interest per annum will Rs. 5,000 double itself in 8 years?

A) 10%
B) 12.5%
C) 15%
D) 16.66%

Q.39. A shopkeeper sells an article for Rs. 840 at a loss of 4%. Find the cost price of the article.

A) Rs. 875
B) Rs. 880
C) Rs. 900
D) Rs. 920

Q.40. In an examination, 85% of the total candidates passed. If 90 candidates failed, find the total number of candidates who appeared.

A) 500
B) 600
C) 750
D) 800

Q.41. The average age of 4 boys is 12 years. If the age of their teacher is included, the average age becomes 15 years. What is the teacher's age?

A) 25 years
B) 27 years
C) 30 years
D) 32 years

Q.42. Divide Rs. 1,050 among A, B, and C in the ratio 2 : 3 : 5. Find B's share.

A) Rs. 210
B) Rs. 315
C) Rs. 525
D) Rs. 350

Q.43. An angle is equal to one-third of its supplement. Find the measure of the angle in degrees.

A) 30 degrees
B) 45 degrees
C) 60 degrees
D) 135 degrees

Q.44. The side of a square hall is 16 m. Find the cost of carpeting it at the rate of Rs. 25 per sq m.

A) Rs. 5,400
B) Rs. 6,000
C) Rs. 6,400
D) Rs. 7,200

Q.45. A rectangular lawn measures 30 m by 20 m. A running track 3 m wide surrounds it externally. Find the area of the track.

A) 336 sq m
B) 340 sq m
C) 356 sq m
D) 360 sq m

Q.46. How many wooden cubical blocks of side 5 cm can be cut from a wooden cuboid measuring 30 cm x 20 cm x 15 cm?

A) 64
B) 72
C) 80
D) 96

Q.47. Find the capacity in liters of a rectangular water tank measuring 2.5 m x 2 m x 1.2 m. (1 cu m = 1,000 liters)

A) 5,000 L
B) 6,000 L
C) 7,500 L
D) 8,000 L

Q.48. The perimeter of an isosceles triangle is 38 cm. If the non-equal base is 10 cm, find the length of each equal side.

A) 12 cm
B) 14 cm
C) 16 cm
D) 18 cm

Q.49. Find the area of a circle whose radius is 14 cm. (Use pi = 22/7)

A) 308 sq cm
B) 616 sq cm
C) 88 sq cm
D) 1,232 sq cm

Q.50. Find the hypotenuse of a right-angled triangle whose two perpendicular sides are 9 cm and 12 cm.

A) 13 cm
B) 15 cm
C) 17 cm
D) 20 cm

Q.51. Solve for x: 5x - 7 = 3x + 11

A) 7
B) 8
C) 9
D) 10

Q.52. Convert 25 m/s into km/h.

A) 72 km/h
B) 80 km/h
C) 90 km/h
D) 100 km/h

Q.53. If 24 pens cost Rs. 360, how many pens can be purchased for Rs. 525?

A) 32 pens
B) 35 pens
C) 38 pens
D) 40 pens

Q.54. Find the measure of an angle which is equal to its complementary angle.

A) 30 degrees
B) 45 degrees
C) 60 degrees
D) 90 degrees

Q.55. Find the height of a triangle whose area is 84 sq cm and base is 14 cm.

A) 10 cm
B) 12 cm
C) 14 cm
D) 16 cm

Q.56. What is the HCF of two consecutive even numbers?

A) 1
B) 2
C) 4
D) Product of the numbers

Q.57. If 40% of a number is 160, find 75% of that number.

A) 250
B) 280
C) 300
D) 320

Q.58. Simplify: (6.25 x 0.8) / 0.25

A) 15
B) 18
C) 20
D) 25

Q.59. The angles of a quadrilateral are in the ratio 1 : 2 : 3 : 4. Find the largest angle in degrees.

A) 108 degrees
B) 120 degrees
C) 144 degrees
D) 160 degrees

Q.60. Find the arithmetic mean of first six multiples of 4.

A) 12
B) 14
C) 16
D) 18

Q.61. A pipe fills a reservoir in 8 hours. What percentage of the reservoir is filled in 3 hours 12 minutes?

A) 35%
B) 40%
C) 45%
D) 50%

Q.62. Find the length of wire needed to fence a circular garden of diameter 28 m twice. (Use pi = 22/7)

A) 88 m
B) 176 m
C) 264 m
D) 352 m

Q.63. A cyclist covers a distance of 15 km in 45 minutes. Find his speed in km/h.

A) 18 km/h
B) 20 km/h
C) 22.5 km/h
D) 25 km/h

Q.64. Find the compound interest on Rs. 10,000 for 2 years at 10% p.a. compounded annually.

A) Rs. 2,000
B) Rs. 2,100
C) Rs. 2,200
D) Rs. 2,400

Q.65. What key term describes a regular polygon whose interior angle is 108 degrees?

A) Pentagon
B) Hexagon
C) Octagon
D) Decagon

Q.66. Find the LCM of 24, 36, and 40.

A) 180
B) 240
C) 360
D) 480

Q.67. Find the lateral surface area of a cuboid measuring 10 cm x 8 cm x 6 cm.

A) 180 sq cm
B) 216 sq cm
C) 240 sq cm
D) 376 sq cm

Q.68. If 4/7 of a vessel is filled with water, what fraction of the vessel is still empty?

A) 2/7
B) 3/7
C) 4/7
D) 5/7

Q.69. Convert 7/25 into a percentage.

A) 21%
B) 24%
C) 28%
D) 32%

Q.70. In a school library, there are 180 storybooks and 120 science books. Find the ratio of science books to storybooks in simplest form.

A) 2 : 3
B) 3 : 2
C) 3 : 5
D) 2 : 5

Q.71. By selling a watch for Rs. 1,440, a dealer gains 20%. What was the cost price of the watch?

A) Rs. 1,150
B) Rs. 1,200
C) Rs. 1,250
D) Rs. 1,300

Q.72. The average score of 5 subjects is 74. If the total score in 4 subjects is 290, find the score in the 5th subject.

A) 70
B) 76
C) 80
D) 84

Q.73. Find the side of an equilateral triangle whose perimeter is equal to the perimeter of a square of side 9 cm.

A) 10 cm
B) 12 cm
C) 14 cm
D) 15 cm

Q.74. Simplify: 50 - 3 x [18 - {14 - (9 - 4)}]

A) 23
B) 26
C) 29
D) 32

Q.75. Find the smallest number by which 180 must be multiplied to make it a perfect square.

A) 2
B) 3
C) 5
D) 6

SECTION C — INTELLIGENCE | 25 Questions x 2 Marks = 50 Marks

Q.76. Complete the letter series: Z, W, T, Q, N, _____

A) K
B) L
C) M
D) J

Q.77. Find the missing term in the series: 2, 6, 12, 20, 30, 42, _____

A) 52
B) 54
C) 56
D) 60

Q.78. If DELHI is coded as 73541 and CALCUTTA as 82589662, how is CALICUT coded?

A) 8251896
B) 8251862
C) 8251396
D) 8251892

Q.79. In a certain code language, FLOWER is coded as EKNVDQ. How is GARDEN coded in that language?

A) FZQCDM
B) FYQCDM
C) FZQBDM
D) FZQCDN

Q.80. Find the odd one out among the following items:

A) Mercury
B) Sodium
C) Copper
D) Iron

Q.81. Complete the analogy: Clock : Time :: Thermometer : _____

A) Heat
B) Temperature
C) Energy
D) Weather

Q.82. Complete the analogy: 9 : 81 :: 12 : _____

A) 108
B) 132
C) 144
D) 156

Q.83. A man pointing to a photograph says, 'The lady in the photograph is the wife of the only son of my father.' How is the lady related to the man?

A) Sister
B) Wife
C) Daughter
D) Mother

Q.84. A person starts from point A and walks 4 km East, then turns left and walks 3 km to reach point B. What is the shortest distance between A and B?

A) 5 km
B) 6 km
C) 7 km
D) 8 km

Q.85. In a queue of 40 people, Priya is 15th from the front end. What is her position from the rear end?

A) 25th
B) 26th
C) 27th
D) 28th

Q.86. If 'P' denotes '+', 'Q' denotes '-', 'R' denotes 'x', and 'S' denotes '/', then evaluate: 18 R 4 S 6 P 7 Q 5

A) 12
B) 14
C) 16
D) 18

Q.87. Which word can be formed using the letters of the word 'CELEBRATION' without altering the sequence order?

A) ACTION
B) CREATE
C) LION
D) RATION

Q.88. Find the missing number in the pattern: 3 5 15 4 6 24 5 7 ?

A) 30
B) 35
C) 40
D) 42

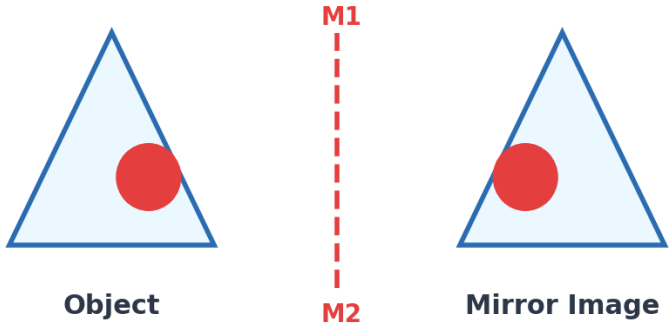
Q.89. Find the missing term: A1Z, C3X, E5V, _____

A) G7T
B) G7U
C) H8T
D) F6U

Q.90. If PEN is coded as 35 and BAT is coded as 23, what is the code for CUP?

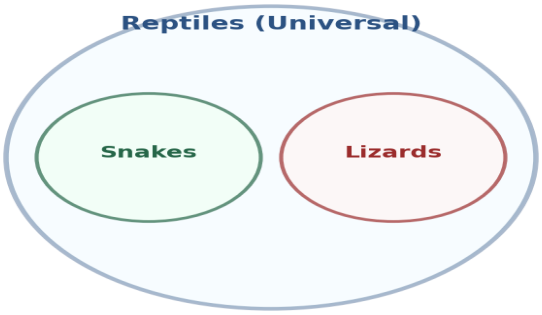
A) 38
B) 40
C) 42
D) 45

Q.91. Choose the correct mirror image of the given geometric figure across the mirror line M1-M2.



A) Mirrored triangle with circle on opposite side
B) Inverted triangle with circle on same side
C) Unchanged figure
D) Upside down figure

Q.92. Which Venn diagram best represents the relationship among 'Reptiles', 'Snakes', and 'Lizards'?



A) Two separate non-overlapping circles inside a large outer circle
B) Three mutually intersecting circles
C) Two overlapping circles inside a third circle
D) Three separate non-overlapping circles

Q.93. A is B's brother. C is A's mother. D is C's father. E is B's son. How is D related to E?

A) Father
B) Grandfather
C) Great-Grandfather
D) Uncle

Q.94. Find the wrong term in the number series: 3, 7, 15, 31, 63, 127, 250

A) 31
B) 63
C) 127
D) 250

Q.95. Six students P, Q, R, S, T, U are standing in a circle facing the center. R is to the immediate left of P. S is between P and Q. T is between R and U. Who is sitting opposite to P?

A) Q
B) R
C) T
D) U

Q.96. If North becomes South-West and East becomes North-West, what will South become?

A) North-East
B) East
C) South-East
D) West

Q.97. How many straight lines are needed at minimum to create the letter 'M'?

A) 2
B) 3
C) 4
D) 5

Q.98. Select the word which is LEAST like the other three:

A) Guitar
B) Violin
C) Flute
D) Cello

Q.99. If 15th August 2023 was a Tuesday, what day of the week was 15th August 2024? (Note: 2024 was a leap year)

A) Wednesday
B) Thursday
C) Friday
D) Saturday

Q.100. Find the missing number in the sequence: 1, 8, 27, 64, 125, _____

A) 196
B) 216
C) 243
D) 256

SECTION D — GENERAL KNOWLEDGE | 25 Questions x 2 Marks = 50 Marks

Q.101. Which instrument is used to record earthquake intensity?

A) Seismograph
B) Lactometer
C) Barometer
D) Odometer

Q.102. Deficiency of which Vitamin leads to Rickets in growing children?

A) Vitamin A
B) Vitamin B
C) Vitamin C
D) Vitamin D

Q.103. Where is the famous Golden Temple located in India?

A) Amritsar
B) Anandpur Sahib
C) Chandigarh
D) Patiala

Q.104. Who was the first Vice-President of Independent India?

A) Dr. S. Radhakrishnan
B) Dr. Zakir Husain
C) V.V. Giri
D) Sardar Patel

Q.105. What is the capital city of Canada?

A) Toronto
B) Vancouver
C) Ottawa
D) Montreal

Q.106. Which part of the human brain controls involuntary body functions like heartbeat and breathing?

A) Cerebrum
B) Cerebellum
C) Medulla Oblongata
D) Spinal Cord

Q.107. Which layer of Earth's atmosphere protects us from harmful ultraviolet (UV) solar radiation?

A) Troposphere
B) Stratosphere (Ozone Layer)
C) Mesosphere
D) Thermosphere

Q.108. In which year did India win its first ICC Cricket World Cup under Kapil Dev's captaincy?

A) 1975
B) 1979
C) 1983
D) 1987

Q.109. Who composed the Indian National Song 'Vande Mataram'?

A) Rabindranath Tagore
B) Bankim Chandra Chattopadhyay
C) Sri Aurobindo
D) Muhammad Iqbal

Q.110. Which Indian state has the longest coastline?

A) Maharashtra
B) Tamil Nadu
C) Gujarat
D) Andhra Pradesh

Q.111. What is the official currency of the United Kingdom?

A) Euro
B) Pound Sterling
C) Dollar
D) Franc

Q.112. Which classical dance form originated in the state of Kerala?

A) Bharatanatyam
B) Kathakali
C) Odissi
D) Kathak

Q.113. Who built the famous Grand Trunk (GT) Road in India?

A) Akbar
B) Sher Shah Suri
C) Shah Jahan
D) Ashoka

Q.114. Which planet is known as the 'Morning Star' or 'Evening Star'?

A) Mercury
B) Venus
C) Mars
D) Jupiter

Q.115. Where is the headquarters of the World Health Organization (WHO) located?

A) New York
B) Geneva
C) Paris
D) London

Q.116. Which national park in Gujarat is the exclusive natural home of Asiatic Lions?

A) Gir National Park
B) Ranthambore National Park
C) Kanha National Park
D) Bandipur National Park

Q.117. How many members can the President of India nominate to the Rajya Sabha?

A) 2
B) 10
C) 12
D) 14

Q.118. Which mineral is essential for the healthy formation of bones and teeth?

A) Iron
B) Calcium
C) Iodine
D) Sodium

Q.119. Which is the largest freshwater lake in India?

A) Chilika Lake
B) Wular Lake
C) Dal Lake
D) Sambhar Lake

Q.120. Who was popularly known as 'Netaji'?

A) Bhagat Singh
B) Subhas Chandra Bose
C) Jawaharlal Nehru
D) Bal Gangadhar Tilak

Q.121. With which sport is the famous 'Santosh Trophy' associated?

A) Cricket
B) Hockey
C) Football
D) Badminton

Q.122. What is the highest wartime gallantry award in India?

A) Maha Vir Chakra
B) Param Vir Chakra
C) Vir Chakra
D) Kirti Chakra

Q.123. Which Indian city is known as the 'Silicon Valley of India'?

A) Hyderabad
B) Pune
C) Bengaluru
D) Chennai

Q.124. Which gas do human beings inhale during respiration?

A) Carbon Dioxide
B) Nitrogen
C) Oxygen
D) Hydrogen

Q.125. Where is the Indian Military Academy (IMA) situated?

A) Dehradun

B) Khadakwasla

C) Chennai

D) Mhow

PART B — MASTER ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	33	A	65	A	97	C
2	B	34	A	66	C	98	C
3	B	35	A	67	B	99	B
4	B	36	B	68	B	100	B
5	B	37	B	69	C	101	A
6	B	38	B	70	A	102	D
7	A	39	A	71	B	103	A
8	D	40	B	72	C	104	A
9	B	41	B	73	B	105	C
10	A	42	B	74	A	106	C
11	A	43	B	75	C	107	B
12	B	44	C	76	A	108	C
13	C	45	A	77	C	109	B
14	A	46	B	78	A	110	C
15	B	47	B	79	A	111	B
16	B	48	B	80	A	112	B
17	B	49	B	81	B	113	B
18	A	50	B	82	C	114	B
19	C	51	C	83	B	115	B
20	B	52	C	84	A	116	A
21	A	53	B	85	B	117	C
22	B	54	B	86	B	118	B
23	B	55	B	87	D	119	B
24	B	56	B	88	B	120	B
25	B	57	C	89	A	121	C
26	D	58	C	90	B	122	B
27	A	59	C	91	A	123	C
28	A	60	B	92	A	124	C
29	B	61	B	93	C	125	A
30	C	62	B	94	D		
31	A	63	B	95	D		
32	C	64	B	96	A		

PART C — STEP-BY-STEP DETAILED SOLUTIONS

SECTION A — LANGUAGE (ENGLISH) — SOLUTIONS

Q.1. Correct Option: (B)

Explanation: The passage states forests absorb carbon dioxide and release oxygen during photosynthesis.

Q.2. Correct Option: (B)

Explanation: Deforestation leads to global warming, unpredictable weather, and loss of biodiversity.

Q.3. Correct Option: (B)

Explanation: 'Van Mahotsav' encourages citizens to plant trees and protect forests.

Q.4. Correct Option: (B)

Explanation: Forests prevent soil erosion and regulate rainwater flow.

Q.5. Correct Option: (B)

Explanation: 'Conservation' (preservation/protection) is the opposite of 'Destruction'.

Q.6. Correct Option: (B)

Explanation: 'Suraj' is a specific person's name and hence a proper noun.

Q.7. Correct Option: (A)

Explanation: 'Ours' is the correct possessive pronoun.

Q.8. Correct Option: (D)

Explanation: 'Costlier' is the comparative form of 'costly'.

Q.9. Correct Option: (B)

Explanation: 'Hour' begins with a silent 'h' and a vowel sound ('our'), so article 'an' is required.

Q.10. Correct Option: (A)

Explanation: 'Behind' indicates position at the back or concealed by the clouds.

Q.11. Correct Option: (A)

Explanation: 'Unless' means 'if not' and fits the conditional negative meaning.

Q.12. Correct Option: (B)

Explanation: 'Hardworking' is a synonym for 'Diligent'.

Q.13. Correct Option: (C)

Explanation: 'Harmful' is the opposite of 'Beneficial'.

Q.14. Correct Option: (A)

Explanation: 'Necessary' is the correct spelling.

Q.15. Correct Option: (B)

Explanation: An 'Autobiography' is written by the person himself/herself.

Q.16. Correct Option: (B)

Explanation: The plural form of 'Ox' is 'Oxen'.

Q.17. Correct Option: (B)

Explanation: 'Nephew' is the masculine counterpart of 'Niece'.

Q.18. Correct Option: (A)

Explanation: A group of stars is called a 'galaxy' or 'constellation'.

Q.19. Correct Option: (C)

Explanation: 'Was playing' indicates Past Continuous Tense.

Q.20. Correct Option: (B)

Explanation: When subjects are joined by 'neither...nor', the verb agrees with the closer subject ('friends' -> plural 'were').

Q.21. Correct Option: (A)

Explanation: 'Wisdom' is an abstract noun representing a quality.

Q.22. Correct Option: (B)

Explanation: Active Simple Past form: 'The postman delivered the letter.'

Q.23. Correct Option: (B)

Explanation: The proverb is 'Actions speak louder than words.'

Q.24. Correct Option: (B)

Explanation: 'To spill the beans' means to reveal secret information.

Q.25. Correct Option: (B)

Explanation: Correct order: 'My grandfather rises early in the morning.'

SECTION B — MATHEMATICS — SOLUTIONS

Q.26. Correct Option: (D)

Explanation: Face value of a digit is the digit itself, so face value of 8 is 8.

Q.27. Correct Option: (A)

Explanation: Place value of first 5 = 5,00,000; Place value of second 5 = 5,000. Sum = 5,05,000.

Q.28. Correct Option: (A)

Explanation: 900 = CM, 40 = XL, 8 = VIII. Total = CMXLVIII.

Q.29. Correct Option: (B)

Explanation: 8,725 rounds to 9,000; 3,480 rounds to 3,000. Estimated difference = 9,000 - 3,000 = 6,000.

Q.30. Correct Option: (C)

Explanation: Parenthesis: 6 x 2 = 12. Braces: 18 - 12 = 6. Square bracket: 12 + 6 = 18. Division: 72 / 18 = 4.

Q.31. Correct Option: (A)

Explanation: LCM of 15, 20, 25: 15=3x5, 20=2^2x5, 25=5^2. LCM = 300. Largest 4-digit = 9999. 9999 / 300 = 33 rem 99. Number = 9999 - 99 = 9,900.

Q.32. Correct Option: (C)

Explanation: The HCF of any two distinct prime numbers is always 1.

Q.33. Correct Option: (A)

Explanation: Convert to improper fractions: 16/3 - 9/4 + 3/2 = (64 - 27 + 18) / 12 = 55/12 = 4 (7/12).

Q.34. Correct Option: (A)

Explanation: Comparing digits left to right: 0.54 > 0.45 > 0.405 > 0.054.

Q.35. Correct Option: (A)

Explanation: 345 x 12 = 4140. 3 decimal places -> 4.14.

Q.36. Correct Option: (B)

Explanation: Speed = 200 / 10 = 20 m/s. Convert to km/h: 20 x (18/5) = 72 km/h.

Q.37. Correct Option: (B)

Explanation: Indirect variation: W1 x D1 = W2 x D2 => 18 x 15 = W2 x 10 => 270 = 10 W2 => W2 = 27 workers.

Q.38. Correct Option: (B)

Explanation: To double itself, Interest = Principal = Rs. 5,000. R = (SI x 100) / (P x T) = (5000 x 100) / (5000 x 8) = 100 / 8 = 12.5%.

Q.39. Correct Option: (A)

Explanation: SP = 96% of CP => 840 = 0.96 x CP => CP = 840 / 0.96 = Rs. 875.

Q.40. Correct Option: (B)

Explanation: Percentage failed = 100% - 85% = 15%. 15% of Total = 90 => Total = (90 x 100) / 15 = 600.

Q.41. Correct Option: (B)

Explanation: Sum of 4 boys' ages = 4 x 12 = 48. Sum of 5 persons' ages = 5 x 15 = 75. Teacher's age = 75 - 48 = 27 years.

Q.42. Correct Option: (B)

Explanation: Sum of ratio parts = 2 + 3 + 5 = 10. Value of 1 part = 1050 / 10 = Rs. 105. B's share = 3 x 105 = Rs. 315.

Q.43. Correct Option: (B)

Explanation: Let angle = x. x = (180 - x) / 3 => 3x = 180 - x => 4x = 180 => x = 45 degrees.

Q.44. Correct Option: (C)

Explanation: Area of square hall = 16 x 16 = 256 sq m. Cost = 256 x 25 = Rs. 6,400.

Q.45. Correct Option: (A)

Explanation: Inner area = 30 x 20 = 600 sq m. Outer dimensions = 36 m by 26 m. Outer area = 36 x 26 = 936 sq m. Area of track = 936 - 600 = 336 sq m.

Q.46. Correct Option: (B)

Explanation: Volume of cuboid = 30 x 20 x 15 = 9,000 cu cm. Volume of 1 cube = 5 x 5 x 5 = 125 cu cm. Number of blocks = 9000 / 125 = 72.

Q.47. Correct Option: (B)

Explanation: Volume = 2.5 x 2 x 1.2 = 6 cu m. Capacity = 6 x 1,000 = 6,000 liters.

Q.48. Correct Option: (B)

Explanation: 2 x Equal side + 10 = 38 => 2 x Equal side = 28 => Equal side = 14 cm.

Q.49. Correct Option: (B)

Explanation: Area = pi x r^2 = (22/7) x 14 x 14 = 22 x 2 x 14 = 616 sq cm.

Q.50. Correct Option: (B)

Explanation: Hypotenuse = sqrt(9^2 + 12^2) = sqrt(81 + 144) = sqrt(225) = 15 cm.

Q.51. Correct Option: (C)

Explanation: 5x - 3x = 11 + 7 => 2x = 18 => x = 9.

Q.52. Correct Option: (C)

Explanation: To convert m/s to km/h, multiply by 18/5: 25 x (18/5) = 90 km/h.

Q.53. Correct Option: (B)

Explanation: Cost of 1 pen = 360 / 24 = Rs. 15. Number of pens for Rs. 525 = 525 / 15 = 35 pens.

Q.54. Correct Option: (B)

Explanation: Let angle = x. x = 90 - x => 2x = 90 => x = 45 degrees.

Q.55. Correct Option: (B)

Explanation: Area = 1/2 x base x height => 84 = 1/2 x 14 x h => 84 = 7 h => h = 12 cm.

Q.56. Correct Option: (B)

Explanation: The HCF of any two consecutive even numbers (e.g., 4 and 6, 12 and 14) is always 2.

Q.57. Correct Option: (C)

Explanation: Number = (160 / 40) x 100 = 400. 75% of 400 = 0.75 x 400 = 300.

Q.58. Correct Option: (C)

Explanation: 6.25 / 0.25 = 25. Then 25 x 0.8 = 20.

Q.59. Correct Option: (C)

Explanation: Sum of angles in quadrilateral = 360 degrees. Ratio sum = 1+2+3+4 = 10. Value of 1 part = 36. Largest angle = 4 x 36 = 144 degrees.

Q.60. Correct Option: (B)

Explanation: First six multiples of 4: 4, 8, 12, 16, 20, 24. Sum = 84. Mean = 84 / 6 = 14.

Q.61. Correct Option: (B)

Explanation: 3 hours 12 mins = 3.2 hours. Percentage = (3.2 / 8) x 100% = 0.4 x 100% = 40%.

Q.62. Correct Option: (B)

Explanation: Circumference = pi x d = (22/7) x 28 = 88 m. Twice fence = 88 x 2 = 176 m.

Q.63. Correct Option: (B)

Explanation: 45 mins = 3/4 hour. Speed = Distance / Time = 15 / (3/4) = 15 x (4/3) = 20 km/h.

Q.64. Correct Option: (B)

Explanation: Amount = 10000 x (1.10)^2 = 10000 x 1.21 = Rs. 12,100. CI = 12100 - 10000 = Rs. 2,100.

Q.65. Correct Option: (A)

Explanation: Formula for interior angle = (n-2) x 180 / n. For n = 5 (Pentagon): 3 x 180 / 5 = 108 degrees.

Q.66. Correct Option: (C)

Explanation: 24 = 2^3 x 3, 36 = 2^2 x 3^2, 40 = 2^3 x 5. LCM = 2^3 x 3^2 x 5 = 8 x 9 x 5 = 360.

Q.67. Correct Option: (B)

Explanation: Lateral surface area = 2 x height x (length + breadth) = 2 x 6 x (10 + 8) = 12 x 18 = 216 sq cm.

Q.68. Correct Option: (B)

Explanation: Fraction empty = 1 - 4/7 = 3/7.

Q.69. Correct Option: (C)

Explanation: (7 / 25) x 100% = 28%.

Q.70. Correct Option: (A)

Explanation: Ratio of science to storybooks = 120 : 180 = 2 : 3.

Q.71. Correct Option: (B)

Explanation: SP = 120% of CP => 1440 = 1.2 x CP => CP = 1440 / 1.2 = Rs. 1,200.

Q.72. Correct Option: (C)

Explanation: Total score in 5 subjects = 5 x 74 = 370. Score in 5th subject = 370 - 290 = 80.

Q.73. Correct Option: (B)

Explanation: Perimeter of square = 4 x 9 = 36 cm. Side of equilateral triangle = 36 / 3 = 12 cm.

Q.74. Correct Option: (A)

Explanation: Inner bracket: 9-4=5. Curly bracket: 14-5=9. Square bracket: 18-9=9. Multiplication: 3 x 9 = 27. Final: 50 - 27 = 23.

Q.75. Correct Option: (C)

Explanation: Prime factorization of 180 = 2^2 x 3^2 x 5. To form pairs, multiply by 5 (180 x 5 = 900 = 30^2).

SECTION C — INTELLIGENCE — SOLUTIONS

Q.76. Correct Option: (A)

Explanation: Reverse alphabetical shift of -3: Z(26) -> W(23) -> T(20) -> Q(17) -> N(14) -> K(11).

Q.77. Correct Option: (C)

Explanation: Differences are consecutive even numbers (+4, +6, +8, +10, +12, +14). Next = 42 + 14 = 56.

Q.78. Correct Option: (A)

Explanation: Direct substitution: C=8, A=2, L=5, I=1, C=8, U=9, T=6 -> 8251896.

Q.79. Correct Option: (A)

Explanation: Each letter is shifted by -1: G->F, A->Z, R->Q, D->C, E->D, N->M (FZQCDM).

Q.80. Correct Option: (A)

Explanation: Mercury is a liquid metal at room temperature, whereas Sodium, Copper, and Iron are solid metals.

Q.81. Correct Option: (B)

Explanation: A Clock measures Time; a Thermometer measures Temperature.

Q.82. Correct Option: (C)

Explanation: Square of the number: 9^2 = 81, 12^2 = 144.

Q.83. Correct Option: (B)

Explanation: 'Only son of my father' means the man himself. The lady is his wife.

Q.84. Correct Option: (A)

Explanation: Shortest distance = sqrt(4^2 + 3^2) = sqrt(16 + 9) = sqrt(25) = 5 km.

Q.85. Correct Option: (B)

Explanation: Position from rear = (Total - Position from front) + 1 = (40 - 15) + 1 = 26th.

Q.86. Correct Option: (B)

Explanation: Substitute symbols: 18 x 4 / 6 + 7 - 5 = 72 / 6 + 7 - 5 = 12 + 7 - 5 = 14.

Q.87. Correct Option: (D)

Explanation: 'RATION' uses letters R, A, T, I, O, N present in 'CELEBRATION'.

Q.88. Correct Option: (B)

Explanation: Row product rule: 3 x 5 = 15; 4 x 6 = 24; 5 x 7 = 35.

Q.89. Correct Option: (A)

Explanation: First letter shifts +2 (A->C->E->G). Number increases +2 (1->3->5->7). Last letter shifts -2 (Z->X->V->T). Result = G7T.

Q.90. Correct Option: (B)

Explanation: Sum of letter positions: C(3) + U(21) + P(16) = 40.

Q.91. Correct Option: (A)

Explanation: Lateral inversion reflects left-right orientation across the vertical mirror line.

Q.92. Correct Option: (A)

Explanation: Both 'Snakes' and 'Lizards' are distinct, non-overlapping groups belonging entirely inside the larger class 'Reptiles'.

Q.93. Correct Option: (C)

Explanation: C is grandmother of E. D is father of C. Therefore, D is the Great-Grandfather of E.

Q.94. Correct Option: (D)

Explanation: Pattern is 2x + 1: 3x2+1=7, 7x2+1=15, 15x2+1=31, 31x2+1=63, 63x2+1=127, 127x2+1=255 (not 250).

Q.95. Correct Option: (D)

Explanation: Circular arrangement clockwise: P, S, Q, U, T, R. Opposite to P is U.

Q.96. Correct Option: (A)

Explanation: Directions rotate 135 degrees counter-clockwise. South rotated 135 degrees counter-clockwise becomes North-East.

Q.97. Correct Option: (C)

Explanation: The letter 'M' consists of 4 straight line segments.

Q.98. Correct Option: (C)

Explanation: Flute is a wind instrument, whereas Guitar, Violin, and Cello are string instruments.

Q.99. Correct Option: (B)

Explanation: 2024 is a leap year, adding 2 odd days between Aug 2023 and Aug 2024. Tuesday + 2 days = Thursday.

Q.100. Correct Option: (B)

Explanation: Cubes of natural numbers: 1^3=1, 2^3=8, 3^3=27, 4^3=64, 5^3=125, 6^3=216.

SECTION D — GENERAL KNOWLEDGE — SOLUTIONS

Q.101. Correct Option: (A)

Explanation: A Seismograph records earthquake waves and intensity.

Q.102. Correct Option: (D)

Explanation: Vitamin D deficiency causes soft, weak bones in children, known as Rickets.

Q.103. Correct Option: (A)

Explanation: The Golden Temple (Sri Harmandir Sahib) is located in Amritsar, Punjab.

Q.104. Correct Option: (A)

Explanation: Dr. Sarvepalli Radhakrishnan served as India's first Vice-President.

Q.105. Correct Option: (C)

Explanation: Ottawa is the official federal capital city of Canada.

Q.106. Correct Option: (C)

Explanation: Medulla Oblongata controls vital involuntary actions like heartbeat and respiration.

Q.107. Correct Option: (B)

Explanation: The Ozone Layer in the Stratosphere absorbs harmful UV rays.

Q.108. Correct Option: (C)

Explanation: India won its first Cricket World Cup on June 25, 1983.

Q.109. Correct Option: (B)

Explanation: Bankim Chandra Chattopadhyay wrote 'Vande Mataram' in his novel Anandamath.

Q.110. Correct Option: (C)

Explanation: Gujarat has the longest mainland coastline in India (approx 1,600 km).

Q.111. Correct Option: (B)

Explanation: The Pound Sterling (GBP) is the official currency of the UK.

Q.112. Correct Option: (B)

Explanation: Kathakali is a major classical dance-drama originating in Kerala.

Q.113. Correct Option: (B)

Explanation: Sher Shah Suri constructed the Grand Trunk Road connecting Bengal to Kabul.

Q.114. Correct Option: (B)

Explanation: Venus shines brightly in the sky and is called the Morning or Evening Star.

Q.115. Correct Option: (B)

Explanation: WHO headquarters is situated in Geneva, Switzerland.

Q.116. Correct Option: (A)

Explanation: Gir National Park in Gujarat is the only natural habitat of Asiatic Lions.

Q.117. Correct Option: (C)

Explanation: The President nominates 12 members to Rajya Sabha for distinguished contributions in art, literature, science, and social service.

Q.118. Correct Option: (B)

Explanation: Calcium is essential for strong bones and teeth.

Q.119. Correct Option: (B)

Explanation: Wular Lake in Jammu and Kashmir is the largest natural freshwater lake in India.

Q.120. Correct Option: (B)

Explanation: Subhas Chandra Bose was affectionately called 'Netaji'.

Q.121. Correct Option: (C)

Explanation: The Santosh Trophy is a premier national football championship in India.

Q.122. Correct Option: (B)

Explanation: Param Vir Chakra is India's highest wartime military decoraton.

Q.123. Correct Option: (C)

Explanation: Bengaluru is called the Silicon Valley of India for its IT industry.

Q.124. Correct Option: (C)

Explanation: Humans inhale oxygen from the air to generate energy during respiration.

Q.125. Correct Option: (A)

Explanation: The Indian Military Academy (IMA) is located in Dehradun, Uttarakhand.